

Disproof of TLMC Conjecture 00000000119

The Pisano period $\pi(p)$ of a prime p is prime only for $p = 2$

TLMC submission 00000000119

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Conjecture 1 (TLMC 00000000119). *There are infinitely many primes p for which the Pisano period $\pi(p)$ (the period of the Fibonacci sequence $F_0 = 0, F_1 = 1, F_{n+2} = F_{n+1} + F_n$ modulo p) is itself prime.*

1 Verdict and disproof

The conjecture is **false**. The set of primes whose Pisano period is prime is exactly $\{2\}$: the referee's own data gives $\pi(2) = 3$ (prime) and $\pi(3) = 8, \pi(5) = 20, \pi(7) = 16$ (all even), and the obstruction generalizes to every prime $p \geq 3$.

Fact 2 (Matrix setup). *Let $Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, so $\det Q = -1$ and $Q^m = \begin{pmatrix} F_{m+1} & F_m \\ F_m & F_{m-1} \end{pmatrix}$.*

Lemma 3 (Parity obstruction). *Let $p \geq 3$ and let $n \geq 1$ satisfy $F_n \equiv 0 \pmod{p}$ and $F_{n+1} \equiv 1 \pmod{p}$. Then n is even and $n \neq 2$.*

Proof. The congruences say $Q^n \equiv I \pmod{p}$, hence $\det(Q^n) = (-1)^n \equiv 1 \pmod{p}$. Since $p \geq 3$, this forces n even. Moreover $n = 2$ is impossible: $F_2 = 1 \not\equiv 0 \pmod{p}$. $\square$

Lemma 4 (Cassini). *For all n , $F_{n+1}^2 - F_n F_{n+2} = (-1)^n$.*

Equivalently, modulo an odd p the period congruences of Lemma 3 give $1 \equiv (-1)^n$, which is absurd for odd n ; this identity is the engine behind the parity statement and is proved in the Lean development ('cassini_even', 'cassini_odd').

Theorem 5. *The Pisano period $\pi(p)$ of a prime p is prime if and only if $p = 2$, in which case $\pi(2) = 3$.*

Proof. $\pi(2) = 3$ is prime (direct computation, 'pisano2' in the Lean file). Conversely let $p \geq 3$ be prime. By Lemma 3 the period $\pi(p)$ is even and different from 2, hence composite. So no prime $p \geq 3$ has prime Pisano period, and the set of the conjecture is exactly $\{2\}$. $\square$

2 Formal verification

The Lean development (directory `lean4/`, core Lean 4 only, no Mathlib) contains a fully axiom-free proof of Theorem 5:

- `period_even`: for every $p \geq 3$ (primality of p is not needed) every period n is even and $\neq 2$;

- `only_two`: a prime period pair (p, n) of primes forces $p = 2$;
- `pisano2, \dots, pisano13`: the referee's numbers $\pi(2)=3, \pi(3)=8, \pi(5)=20, \pi(7)=16, \pi(11)=10, \pi(13)=28$, each a single `rfl` computation that also verifies minimality and the wrap-around.

Every declaration is audited by `Check.lean` with `#print axioms` and depends on *no* axioms. To keep the audit clean the development avoids `omega`, `simp` and `decide` on symbolic goals; all congruences are phrased existentially.

3 Independent recomputation

`reproduce.py` recomputes $\pi(p)$ by fast doubling for the referee's data points, verifies that the only prime below 2000 with prime Pisano period is 2, and checks that all other primes below 2000 have even $\pi(p)$.

4 Scope

The disproof is unconditional: no variant of the conjecture survives.